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Structural and Spectroscopic Characterization of Plutonium and Other Tetravalent Metals Complexed to a Keggin Ion

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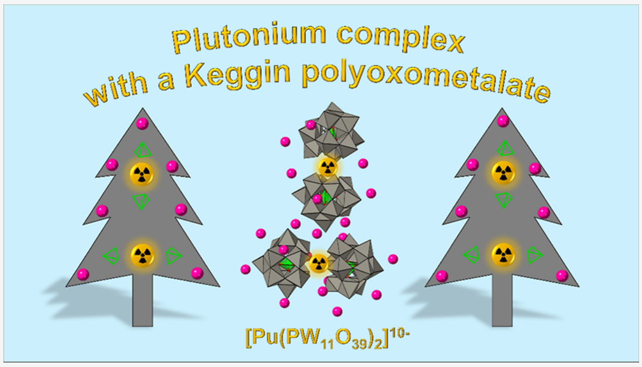
ABSTRACT: We report the isolation of the first plutonium(IV) complex with a Keggin ion chelator: $Cs_{20}[Pu(PW_{11}O_{39})_2]_2\cdot 13H_2O$. Single crystal XRD and solid-state UV−vis absorbance analysis demonstrate the stabilization of Pu^{4+} by the Keggin ligand. The unit cell contains two $[Pu(PW_{11}O_{39})_2]^{10-}$ complexes ($Pu(PW_{11})_2$) bridged by Cs^+. Raman and ^{31}P NMR spectra of $Pu(PW_{11})_2$ are consistent with the analogous Zr^{4+}, Hf^{4+}, Ce^{4+}, and Th^{4+} complexes. The Pu−O bond distances at the two Pu sites are 2.35(3) and 2.34(3) Å, matching the value extrapolated from the bonding trend built with the other 8-coordinated tetravalent cations. However, the long-range arrangement of the $Pu(PW_{11})_2$ complexes within the lattice is unique in the series of $M^{IV}(PW_{11})_2$ compounds: pairs of $Pu(PW_{11})_2$ are organized perpendicular to each other. Based on solution-state UV−visible absorbance, small-angle X-ray scattering (SAXS), and ^{31}P NMR, the tetravalent cations quantitatively form the 1:2 species in solution ($[Pu(PW_{11}O_{39})_2]^{10-}{}_{(aq)}$) and no 1:1 species ($[Pu(PW_{11}O_{39})(H_2O)_x]^{3-}{}_{(aq)}$). Finally, a linear correlation exists between the metal−oxygen distances in the $M^{IV}(PW_{11})_2$ compounds and the corresponding metal dioxides, allowing for extrapolation for Pa^{4+}, Am^{4+}, and Bk^{4+}. The results indicate that our microscale POM approach represents a viable pathway to probe properties of rare actinide ions in discrete molecules, beyond the traditional oxide extended solids.



INTRODUCTION

Plutonium exhibits one of the most diverse chemistries within the periodic table, capable of forming a wide range of compounds spanning oxidation states from +II to +VII. Since its discovery in 1940 by Seaborg and co-workers,[1] many different classes of Pu compounds have been synthesized and characterized (structurally and/or spectroscopically). Besides its numerous alloys and metallic phases,[2] a few hundred structures of coordination compounds that encompass Pu cations are currently known. Pu compounds have been isolated as oxides,[3] nanoparticles or oxo/hydroxo-clusters,[4−7] complexes with inorganic ions,[8−13] organometallics,[14−16] complexes with small organic ligands[17−19] or proteins,[20−23] Pu-based metal organic frameworks,[24,25] and various other chemical systems.[26] As of July 2025, the Inorganic Crystal Structure Database (ICSD)[27] lists 660 inorganic compounds with at least one Pu ion, while the Cambridge Crystallographic Data Centre (CCDC)[28] lists 242 compounds of Pu (mostly materials containing one or more carbon-containing small molecules). Despite the relatively well-documented literature on Pu materials, and the diversity of plutonium compounds and structures reported so far (in terms of ligand types, Pu-oxidation states, and coordination geometries), very few of them are with polyoxometalate (POM) ligands. In fact, prior to this study, only five Pu−POM compounds had been isolated, representing <1% of the known Pu compounds. Table 1 summarizes the Pu−POM compounds that have been synthesized and structurally characterized to date. The only Pu−POM compounds isolated prior to this study were two with the plutonyl(VI) ion, PuO_2^{2+}, and three with the plutonium(IV) ion, Pu^{4+}. No POM compound with other oxidation states of Pu has been reported so far. It is worth noting that Antonio and Chiang[29] reported in 2008 the incorporation of plutonium(III) into the POM $[NaP_5W_{30}O_{110}]^{15-}$ (commonly known as a "Preyssler ion"), but the resulting Pu complex was not structurally characterized.

The year 2009 saw the first crystal structures of Pu−POM compounds being reported, with Sokolova et al.[32] successfully crystallizing a sandwich complex of Pu^{4+} with two $P_2W_{17}O_{61}^{10-}$ ligands, while Copping et al.[34] isolated a polynuclear complex of PuO_2^{2+} with the $GeW_9O_{34}^{10-}$ ligand. These two polyoxotungstate POMs correspond to a monolacunary "Wells−Dawson"

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Table 1. Plutonium(IV) Complexes with POMs That Have Been Structurally Characterized So Far[a]

cation	POM ligand	compound formula	year	ref
Pu^{4+}	$PW_{11}O_{39}^{7-}$	$Cs_{20}[Pu(PW_{11}O_{39})_2]_2\cdot 13H_2O$	2025	this work
	$W_5O_{18}^{6-}$	$Cs_8Pu(W_5O_{18})_2\cdot CsCl\cdot 6.5H_2O$	2025	30
	$Mo_{12}O_{42}^{12-}$	$Na_7H[PuMo_{12}O_{42}]\cdot 12H_2O$	2016	31
	$P_2W_{17}O_{61}^{10-}$	$K_{12}H_4Pu(P_2W_{17}O_{61})_2\cdot 19H_2O$	2009	32
PuO_2^{2+}	$Se_6W_{45}O_{159}^{24-}$	$[(CH_3)_2NH_2]_x(H_3O)_{22-x}[(PuO_2)Se_6W_{45}O_{159}(H_2O)_{10}]\cdot nH_2O$	2023	33
	$GeW_9O_{34}^{10-}$	$K_{11}[K_3(PuO_2)_3(GeW_9O_{34})_2]\cdot 12H_2O$	2009	34

[a]For the sake of comparison, the two plutonyl(VI)−POM complexes that have been reported are also included in this compilation. No Pu(II), Pu(III), Pu(V), or Pu(VII)−POM compound has been structurally characterized so far. We also note that a Pu(III) tungstate compound, $PuW_2O_7(OH)(H_2O)$, has been reported by Cross et al.,[12] but this is not a POM compound. Pu(III) was also studied with the Preyssler-type POM, $[NaP_5W_{30}O_{110}]^{14-}$, by Antonio and Chiang,[29] but the complex has not been structurally characterized.

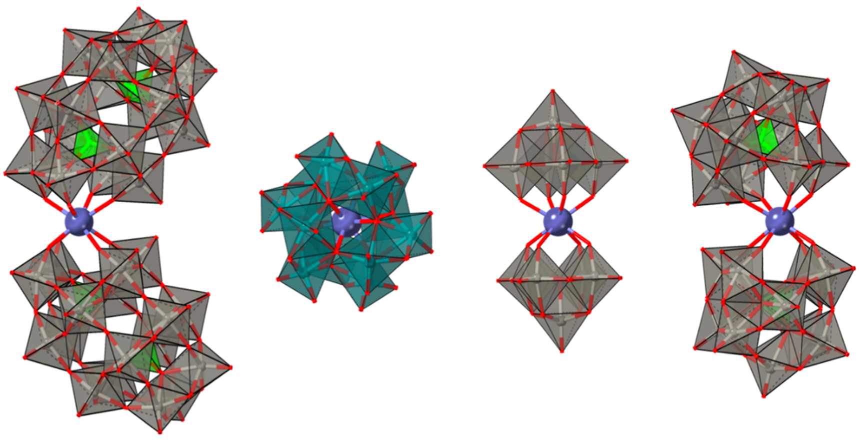

Figure 1. Summary of the polyoxometalate complexes with plutonium(IV) isolated thus far. Complexes are displayed at the same scale for general size comparison. Color code: purple (Pu), gray (W), cyan (Mo), red (oxygen), and green (phosphorus). Counterions and water molecules are omitted for clarity. See Table 1 for references and full compound formulas.

complex[35] and a trilacunary "Keggin" complex,[36] respectively, which are two of the main categories of POMs (Figure 1). In 2016, Charushnikova et al.[31] expanded the chemistry of Pu−POM to polyoxomolybdate chemistry, with the structural characterization of Pu^{4+} encapsulated in $Mo_{12}O_{42}^{12-}$, also called a "Dexter−Silverton" POM.[37] In 2023, Zhang et al.[33] reported the fourth, and thus far largest, Pu−POM complex. This particular compound encompasses one PuO_2^{2+} cation bound to one large (∼20 Å) and lacunary selenopolytungstate anion $Se_6W_{45}O_{159}^{24-}$. Finally, our team recently[30] isolated a Pu^{4+} compound with one of the smallest polytungstate anions that can be made, the $W_5O_{18}^{6-}$ ligand, forming the complex $[Pu(W_5O_{18})_2]^{8-}$. This type of complex belongs to the "Peacock−Weakley" POM category.[38] We note that the "Peacock−Weakley" complexes are often mischaracterized as "Lindqvist" ions in the literature even though these two structures are readily different as the Lindqvist structure is nonlacunary.

Thus far, one major class of POMs ligands, namely, the monolacunary Keggin[36] ions (i.e., $XW_{11}O_{39}^{n-}$), has remained absent from the chemistry of Pu. This type of ligand is perhaps the most studied class of POMs, and it can form complexes with almost any element in the periodic table. However, its complexes with transuranic elements have remained elusive mainly because the synthetic protocols previously used were not compatible with radiochemistry experiments. In the present work, we address this gap in the literature by reporting the synthesis and characterization of the first Pu^{4+} complex with a Keggin-type ligand (Figure 1). The compound, fully formulated as $Cs_{20}[Pu(PW_{11}O_{39})_2]_2\cdot 13H_2O$, was synthesized at the microscale,[39] its single crystal structure was determined, and its spectroscopic properties were investigated via Raman microscopy plus UV−visible spectrophotometry in both solution and the solid state, as well as solution-state ^{31}P NMR. Due to radiological limitations inherent to plutonium samples, analogous complexes with Zr^{4+}, Hf^{4+}, Ce^{4+}, and Th^{4+} were probed via solution-state SAXS, to investigate $[M(PW_{11}O_{39})_2]^{10-}$ speciation in solution.

■ RESULTS

Spectroscopic Characterization of the Pu(IV) and Other Tetravalent Ion Complexes with the POM Ligand $PW_{11}O_{39}^{7-}$. This study of $M^{IV}(PW_{11})_2$ compounds was initiated with Ce(IV) as a proxy for Pu(IV) to evaluate the ability of the $PW_{11}O_{39}^{7-}$ ligands to mitigate the propensity of tetravalent cations toward hydrolysis in aqueous solution, which often results in colloidal PuO_{2+x}.[40,41] As shown in Figures S1 and S2, Ce(IV) can be maintained in aqueous solution upon binding to the $PW_{11}O_{39}^{7-}$ ligand, even when using just the stoichiometric ratio 1:2 Ce:POM. Aqueous solutions of $Ce^{IV}(PW_{11})_2$ remain stable under ambient conditions for months, even at pH ∼5 (based on UV absorbance characteristics). As we have been able to crystallize the $Ce^{IV}(PW_{11})_2$ complex (obtained as $Cs_{10}Ce(PW_{11}O_{39})_2\cdot 5H_2O$),[42] these results encouraged us to pursue the

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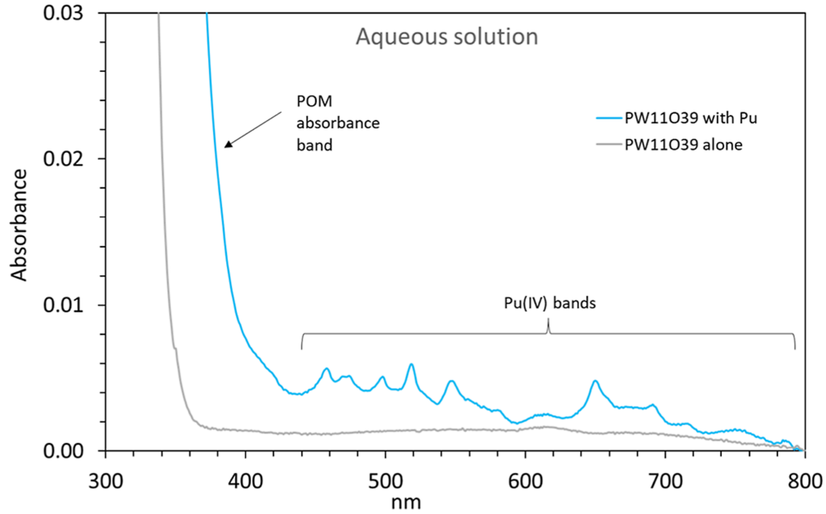

Figure 2. Absorbance spectra of aqueous solutions of the $PW_{11}O_{39}^{7-}$ ligand before (gray curve) and after (blue curve) addition of Pu(IV). Buffer: 0.5 M $NaCH_3COO$, pH 4.5. [Pu] = 100 μM. [POM] = 200 μM. Additional spectra are given in Figure S3.

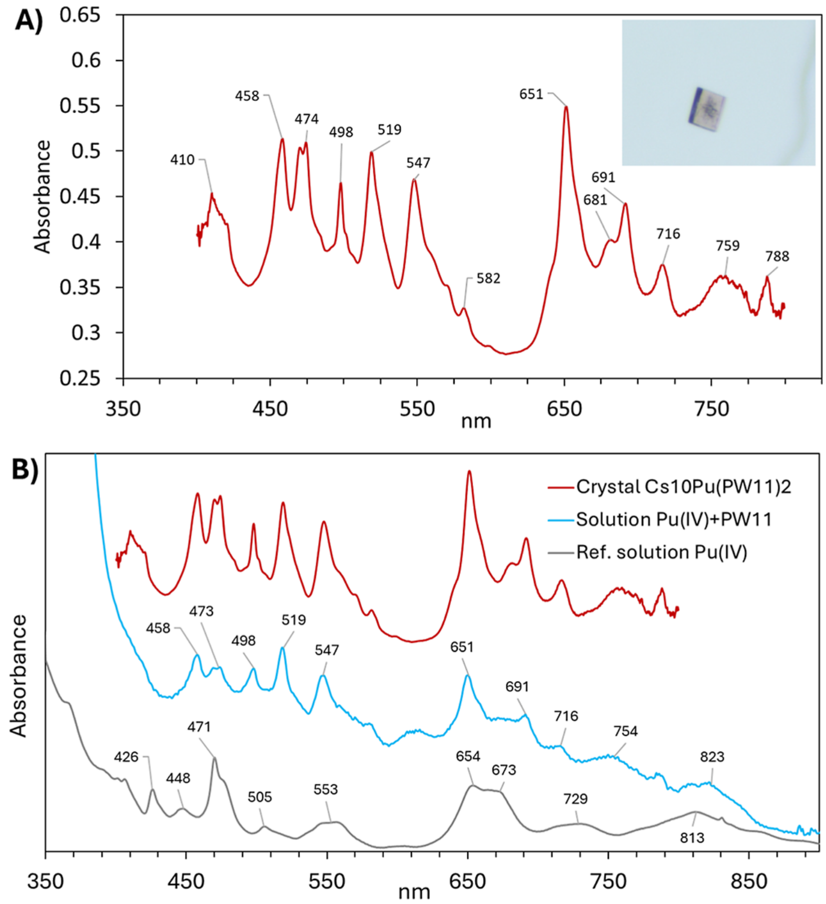

Figure 3. (A) Solid-state UV−vis absorbance spectrum of the Pu(IV) complex with PW_{11}. The spectrum is the average of 600 individual spectra, collected on 6 different crystals (100 per crystal). Inset: Picture of a crystal used for analysis. The size of the crystal is about 50 × 40 × 10 μm. (B) Solid-state (single crystal—top spectrum) and solution-state (aqueous solution—middle spectrum) UV−vis absorbance of plutonium in the presence of 2 equiv of PW_{11}. The bottom spectrum is for a Pu(IV) solution in 2 M HCl, shown here for reference.

characterization of the analogous compound with plutonium, tentatively the $[Pu(PW_{11}O_{39})_2]^{10-}$ complex (hereafter, $Pu(PW_{11})_2$). Figure 2 shows the UV−visible absorbance of an aqueous solution of $PW_{11}O_{39}^{7-}$ to which a stoichiometric amount of Pu(IV) was added (POM:Pu ratio of 2:1). The sample was buffered at pH 4.5—conditions under which Pu(IV) is not stable in aqueous solution but rather undergoes redox reactions or hydrolyzes to form hydroxides, colloids, or precipitates as Pu oxides, in the absence of a ligand such as the POM.

The absorbance spectra of $Pu(PW_{11})_2$ solutions (Figures 2 and 3) exhibit the characteristic absorbance bands of the +IV

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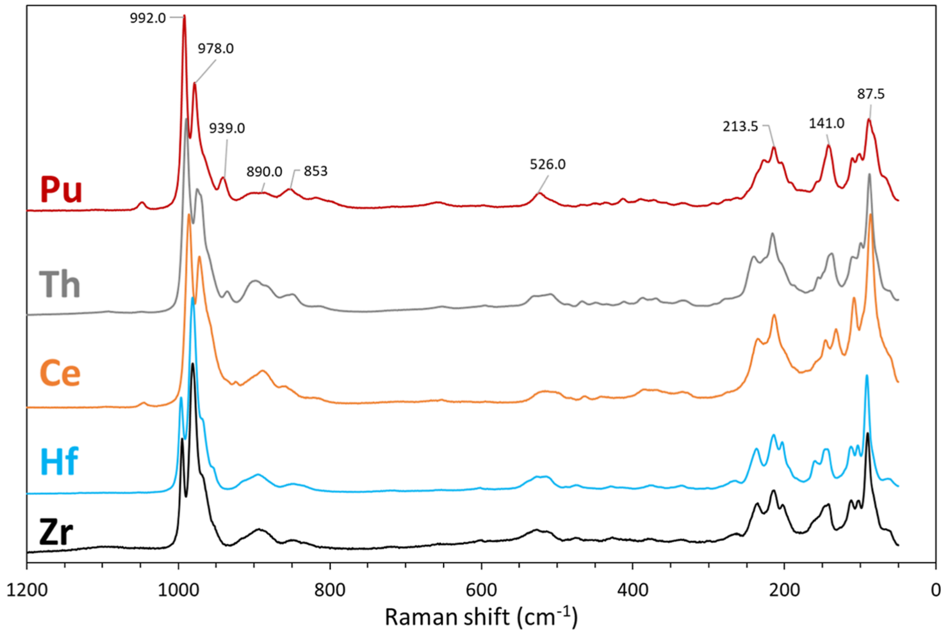

Figure 4. Solid-state Raman spectrum of the plutonium(IV) complex with the Keggin POM ligand ($PW_{11}O_{39}^{7-}$) and comparison with the Th(IV), Ce(IV), Hf(IV), and Zr(IV) analogous compounds. Additional detailed views of the individual spectra are given in Figures S5−S9.

oxidation state[2] of Pu and show no evidence of other Pu-oxidation states. When compared to the absorbance of the unbound Pu(IV) ion (reference spectrum in Figure 3B), all the Pu(IV) bands are observed, except for the one at ∼426 nm because it is at the threshold of the POM's own absorbance. The $PW_{11}O_{39}^{7-}$ absorbance band (due to W−O bonds) is typically at 350 nm but shifts toward higher wavelengths, by about 40 nm, upon forming the $Pu(PW_{11})_2$ complex. This bathochromic shift is ∼4 times larger than that observed for the corresponding Ce(IV) complex (Figures S1 and S2).

Upon addition of excess CsCl to solutions of Pu(IV) containing two equivalents of $PW_{11}O_{39}^{7-}$, single crystals with a light red color appear after about 5 days at room temperature (Figures 3A and S4). As we noticed that none of the previously reported protocols for Keggin POM compounds would be compatible with Pu chemistry (due to the high quantity of radioisotope needed), we performed the crystallization tests for $Pu(PW_{11})_2$ at the microscale, using 6 μg of weapons-grade Pu. We previously described this scale-minimization approach for other POM complexes with transuranic elements.[30,39,43,44] The minimization of the actinide quantity needed to synthesize the POM complex allowed us to directly use weapons-grade Pu (94% ^{239}Pu, 6% ^{240}Pu)—a material whose specific activity is ∼16 to ∼3300 times higher than the typical plutonium research isotopes (i.e., ^{242}Pu and ^{244}Pu) but that is less expensive and less rare. The synthesis of $Pu(PW_{11})_2$ was repeated at least three times, with consistent results based on the crystal morphology and their spectroscopic properties (Raman and UV−vis absorbance).

To confirm the synthesis of a Keggin POM complex of Pu(IV), the obtained crystals were first characterized via Raman microscopy. We also synthesized analogous compounds with Zr(IV), Hf(IV), Ce(IV), and Th(IV) as a benchmark. As shown in Figures 4 and S5−S9, the Raman spectrum of $Pu(PW_{11})_2$ is consistent with that of the other tetravalent metal complexes with $PW_{11}O_{39}^{7-}$. The Raman emission pattern of $Pu(PW_{11})_2$ is nearly a perfect match for that of the two other tetravalent f-element complexes tested, $Cs_{10}Ce(PW_{11}O_{39})_2{\cdot}5H_2O$ and $Cs_{10}Th(PW_{11}O_{39})_2{\cdot}5H_2O$, with the characteristic of terminal $\nu(W{=}O_t)$ stretching modes in the 940−1000 cm^{-1} region, the ν(W−O−W) bridging and ν(W−O−Pu) stretches at 500−850 cm^{-1}, the bending, rocking, and twisting modes of the complex around 300−450 cm^{-1}, and Pu−O stretches around 210 cm^{-1}.[45] We tentatively ascribe the lower wavenumber peaks <150 cm^{-1} to the Pu−O−Pu deformations.

As displayed in Figure 3B, the solid-state absorbance spectra of the $Pu(PW_{11})_2$ crystals and that of solutions of Pu(IV) plus PW_{11} are nearly identical. This indicates that the speciation of the Pu(IV)−PW_{11} system remains the same in aqueous solution and in the solid state. Knowing that the solid-state speciation corresponds to the 1:2 complex $Pu(PW_{11})_2$ (see crystallography results below), this suggests that only the 1:2 complex $Pu(PW_{11})_2$ is present in solution with no 1:1 complex or other species. This behavior departs from that of the lanthanide(III)−PW_{11} system, as well as Am(III)−PW_{11} and Cm(III)−PW_{11} systems, which forms almost an equivalent mixture of 1:1 and 1:2 complexes (i.e., $[M^{III}PW_{11}O_{39}(H_2O)_n]^{4-}$ and $[M^{III}(PW_{11}O_{39})_2]^{11-}$) in solution under similar conditions, based on NMR and UV−vis analysis.[39,46] We hypothesize that the 1:2 complex with Pu(IV) is favored over the 1:1 complex in solution due to the higher charge to ionic radius of the tetravalent cations, which drives the speciation toward the more stable 1:2 complex. The formation of only the 1:2 complex $Pu(PW_{11})_2$ in solution is also in line with a recent study that we performed on the equivalent neptunium(IV) system.[47] The hypothetical formation of $[Pu^{IV}PW_{11}O_{39}(H_2O)_n]^{3-}$ would also leave the plutonium ion exposed to hydrolysis due to the multiple water molecules in its first coordination sphere, which we do not observe.

In order to confirm the solution speciation of Pu in the presence of PW_{11}, we also collected the ^{31}P NMR spectrum of a sample containing Pu(IV) and two equivalents of PW_{11}. We also recently reported[47] equivalent NMR experiments for samples with Zr(IV), Hf(IV), and Th(IV). As shown in Figure S10, the ^{31}P NMR spectrum of the Pu(IV) sample presents characteristics consistent with the other tetravalent metal complexes. The

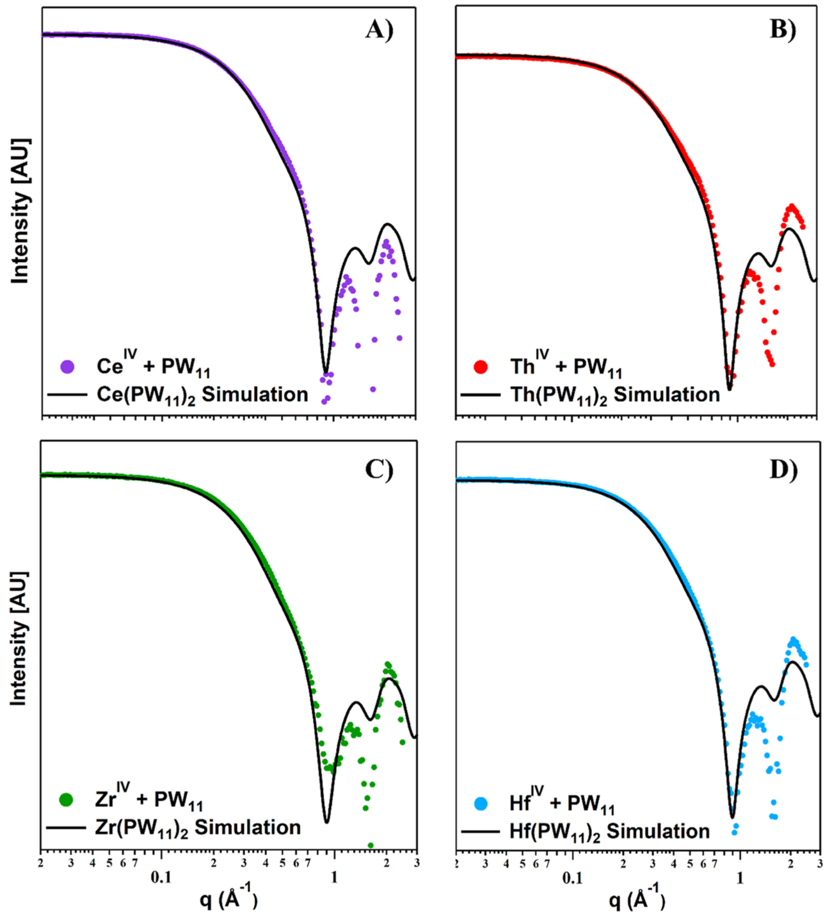

Figure 5. SAXS for solutions of PW_{11} plus tetravalent metals in a 2:1 ratio, compared to simulated $M(PW_{11})_2$ scattering for M = (A) Ce(IV), (B) Th(IV), (C) Zr(IV), and (D) Hf(IV). [Metal] = 5 mM. [POM] = 10 mM. pH = 5 (0.1 M $NaCH_3COO$).

solution-state ^{31}P NMR spectrum of $Pu(PW_{11})_2$ exhibits two resonances, at −21.5 and −23.2 ppm. The resonances for $Pu(PW_{11})_2$ are shifted upfield by 10−12 ppm compared to $Zr(PW_{11})_2$, $Hf(PW_{11})_2$, and $Th(PW_{11})_2$ which is consistent with the paramagnetic shifts stemming from the paramagnetism of Pu(IV). On the other hand, Zr(IV), Hf(IV), and Th(IV) are diamagnetic ions and the resultant chemical shifts have no paramagnetic contributions and therefore have resonances closer to that of the free POM. Taken together, the UV−vis absorbance and NMR results are consistent with the formation of the 1:2 complex in solution, i.e., $[Pu(PW_{11}O_{39})_2]^{10-}$. The clean and invariable formation of the 1:2 complex in solutions of tetravalent cations with PW_{11} was further confirmed via SAXS measurements, as detailed below.

Solution-State SAXS of Analogous Complexes of $Pu(PW_{11}O_{39})_2$. Small-angle X-ray scattering data of PW_{11} solutions mixed in a 2:1 ratio with tetravalent cations (5 mM of Ce^{4+}, Th^{4+}, Zr^{4+}, or Hf^{4+}; 10 mM of PW_{11}, see experimental section for SAXS data collection information) were compared to simulated scattering curves[48] of $M(PW_{11})_2$. As shown in Figure 5, the agreement between simulated and experimental scattering in the Guinier region (0.09 to 0.7 $Å^{-1}$) is excellent, confirming dominant complexation of the tetravalent metals to the POM in a 1:2 ratio in solutions. The similarity between experimental and simulated scattering throughout the entire measured q-range, especially observation of the two oscillations in the high q-region (1 to 2.5 $Å^{-1}$), indicates purity and monospecificity of the solutions, while the misfit in the oscillatory region is due to solvent scattering, which is inherently impossible to remove entirely. Further evidence of the solution speciation is observed in the pair distance distribution function (PDDF) analysis (Figure S11); a real-space representation of scattering data as a histogram of probability as a function of dimension of all scattering vectors through the particles.[49] The shape of the PDDFs features two Gaussian peaks, with the second being less intense than the first. This shape is characteristic of a 1:2 complex[50] in solution. The radius of gyration (R_g, a shape-independent root-mean-square of weighted distances from the center of mass) of the clusters in solution obtained from PDDF modeling ranges from 5.7 Å to 5.9 Å, which matches simulations for $M(PW_{11})_2$ species (Figure S11). Simulation of $[PW_{12}O_{40}]^{3-}$ (PW_{12}) and $[P_2W_{18}O_{62}]^{6-}$ (P_2W_{18}), additional possible solution species, is compared to experimental scattering in Figure S12. Both of these POMs exhibit very different scattering profiles compared to the $M(PW_{11})_2$ complexes and to each other, suggesting that these species are minimally present in solution, further highlighting the strong complexation of lacunary POMs to the tetravalent cations.

Structural Characterization of the Pu(IV) Complex with $PW_{11}O_{39}^{7-}$. Single crystal XRD analysis confirmed the isolation of a Pu(IV) complex with the Keggin ligand of $PW_{11}O_{39}^{7-}$. The compound crystallized as $Cs_{20}[Pu(PW_{11}O_{39})_2]_2 \cdot 13H_2O$ (hereafter $Cs_{10}Pu(PW_{11})_2$), with two crystallographically unique $[Pu(PW_{11}O_{39})_2]^{10-}$ complexes in the unit cell. It crystallizes in the triclinic crystal system ($P\bar{1}$ space group), with a cell volume of 9375.5 $Å^3$ (see Table S1 full

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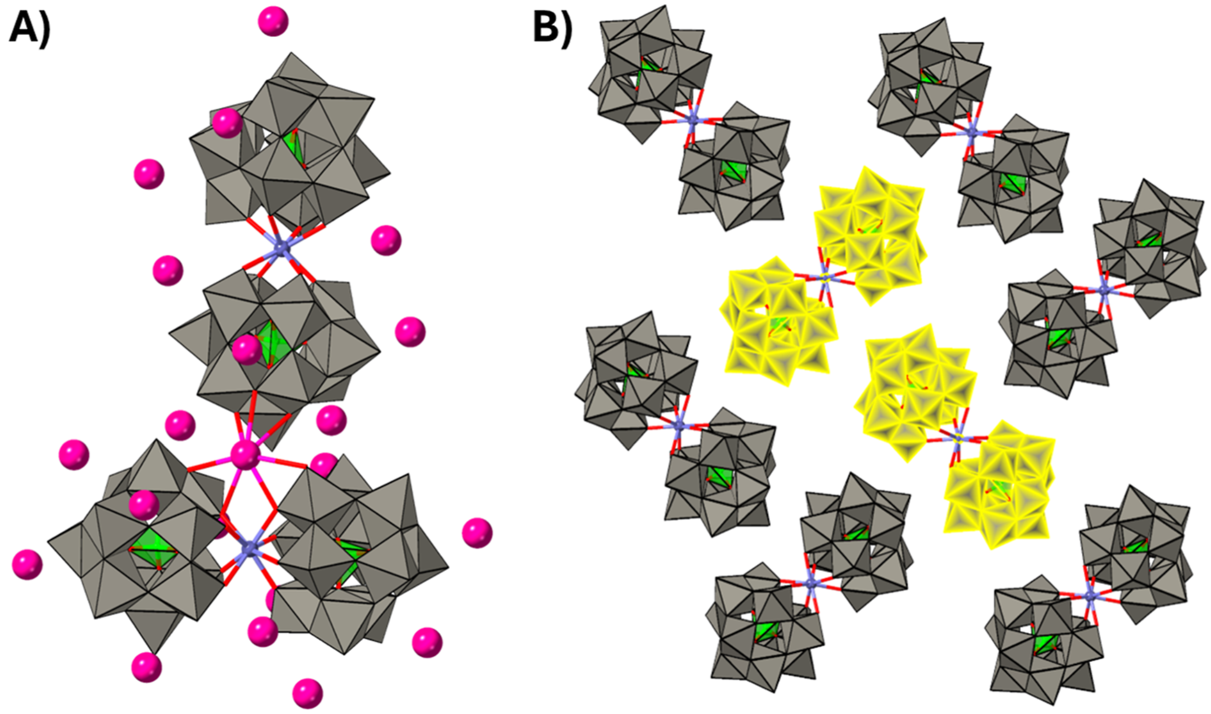

Figure 6. Structure of $Cs_{20}[Pu(PW_{11}O_{39})_2]_2{\cdot}13H_2O$. (A) Arrangement of the complexes in the unit cell. The two individual $Pu(PW_{11})_2$ complexes found in the unit cell are perpendicular to each other. A Cs^+ counterion is located between the two complexes, with a distance Pu−Cs of 4.448 Å (See text). (B) Longer range arrangement of the complexes, with the unitary motif highlighted in yellow. For clarity, the Cs^+ counterions are omitted in this view. Color code: purple (Pu), gray (W), pink (Cs), red (oxygen), and green (phosphorus).

Table 2. Comparison of the Coordination of Plutonium in the Three Pu(IV) Polyoxotungstate Compounds That Are Currently Known

Pu−POM complex	$[Pu(W_5O_{18})_2]^{8-}$	$[Pu(PW_{11}O_{39})_2]^{10-}$	$[Pu(P_2W_{17}O_{61})_2]^{16-}$
Pu^{4+} coordination	eight oxygens (two Pu sites)	eight oxygens (two Pu sites)	eight oxygens (one Pu site)
average bond ⟨Pu−O⟩	2.38(2) Å	2.35(3) Å	2.32(5) Å
	2.38(2) Å	2.34(3) Å	
alkali counterions close to Pu	five Cs^+ at 4.35−4.67 Å	three Cs^+ at 4.44−5.09 Å	one K^+ at 4.37 Å
overall bending angle of the complex[a]	178°	162°	132°
polyhedral distortion index[b] (Pu−O)[53]	0.0084	0.0121	0.0158
	0.0080	0.0088	
continuous symmetry operation measure (CSoM)[54]	0.062	0.673	0.447
	0.061	0.485	
complex molecular weight[c]	2653 g/mol	5593 g/mol	8565 g/mol

[a]For $[Pu(W_5O_{18})_2]^{8-}$, we considered the angle W−Pu−W forms between the two outmost W and Pu. For $[Pu(PW_{11}O_{39})_2]^{10-}$, we considered the angle P−Pu−P. For $[Pu(P_2W_{17}O_{61})_2]^{16-}$, we considered the angle P−Pu−P, with the two outmost P. [b]The polyhedral distortion index, PDI (Pu−O), was defined by Baur.[53] It measures the distortion in terms of the variation in the centroid-to-vertex bond distances ($Pu{-}O_i$), relative to the average bond distance, ⟨Pu−O⟩. It is calculated as follows: PDI (Pu−O) = $(\Sigma i = 1, n|Pu{-}O_i - \langle Pu{-}O\rangle|)/(n\langle Pu{-}O\rangle)$. Here, $n = 8$. [c]See the recent publication by Nielsen et al.[54] for definition and more information about the application of CSoM to f-element complexes. Used 239 g/mol for Pu.

crystallographic details). As the Pu(IV) and Ce(IV) ions have nearly identical ionic radius (0.96 Å for Pu, 0.97 for Ce),[51] we had anticipated similar structures. However, while the analogous Ce(IV) compound, namely, $Cs_{10}Ce(PW_{11}O_{39})_2{\cdot}5H_2O$, also crystallizes in the $P\overline{1}$ space group,[42] the cell volume of $Cs_{10}Pu(PW_{11})_2$ is about twice bigger (9376 vs 4661 Å^3). The asymmetric unit of $Cs_{10}Pu(PW_{11})_2$ is best described as two unique $[Pu(PW_{11}O_{39})_2]^{10-}$ complexes arranged in a nearly perfect perpendicular fashion (Figure 6A,B). This type of arrangement is different from any of the analogous structures with tetravalent Zr (Figure S13), Hf (Figure S14), Ce (Figure S15), and Th (Figure S16). Only Pu(IV) leads to a perpendicular arrangement of the complexes in the lattices (Figure S17) while Zr, Hf, and Ce form more compact unit cells with their complexes all arranged in a parallel fashion or parallel pairs with a non-right angle between pairs in the case of Th.

The two Pu sites in $Cs_{20}[Pu(PW_{11}O_{39})_2]_2{\cdot}13H_2O$ feature eight binding oxygens surrounding the actinide in an antiprismatic arrangement. The average Pu−O bond lengths are 2.34(3) and 2.35(3) Å for the two sites. These bond lengths are consistent with those of a Pu(IV) complex. Bond valence sum calculations[52] yield 3.81 and 3.97 for the two Pu sites, consistent with the expected oxidation state (Table S2). The two $Pu(PW_{11})_2$ complexes are slightly bent, with a P−Pu−P angle of 161.1 and 163.3°. Similarly slight deviations from linearity are observed for the Ce, Th, and Np complexes (162.5, 162.8, and 162.5°, respectively), while the analogous complexes with the smaller ions Zr(IV) and Hf(IV) are more linear, with

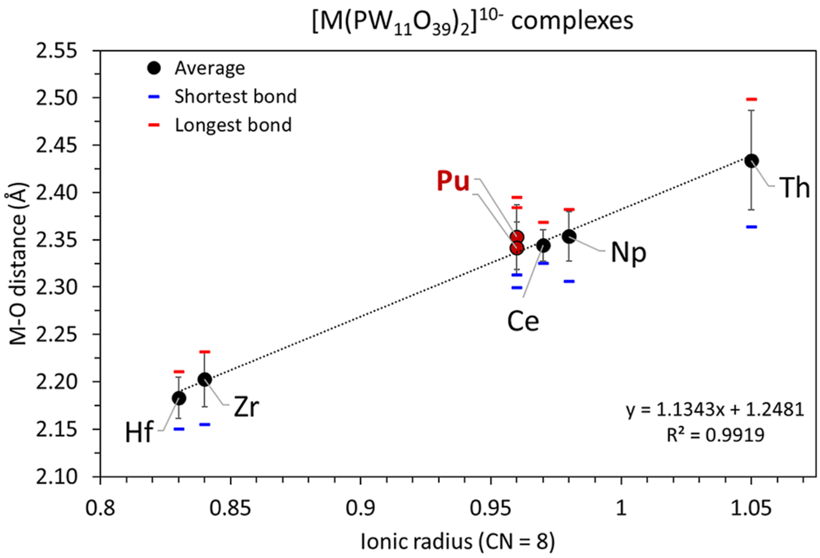

Figure 7. Comparison of the metal−oxygen bond distances observed for the series of tetravalent metals complexed to the Keggin POM $PW_{11}O_{39}{}^{7-}$. All compounds were characterized via single crystal XRD and from isostructures: $Cs_{10}M^{IV}(PW_{11}O_{39})_2{\cdot}nH_2O$. We previously reported the analogous compounds with Hf^{4+}, Zr^{4+}, Ce^{4+}, Th^{4+}, and Np^{4+} elsewhere.[42,47] Two Pu sites are present in the unit cell of the POM complex, with an average Pu−O bond length of 2.353 Å and 2.341 Å. The shortest Pu−O bond lengths for each site are 2.313 Å and 2.300 Å. The longest Pu−O bond lengths are 2.395 Å and 2.384 Å.

both a P−M−P angle of 167.2°. In the $Cs_{10}Pu(PW_{11})_2$ structure, the distance between the two Pu(IV) ions is 14.6 Å. For comparison, the shortest Ce−Ce or Th−Th distances in the $Cs_{10}Ce(PW_{11})_2$ and $Cs_{10}Th(PW_{11})_2$ structures are 10.9 and 11.3 Å, respectively. This metric is a direct effect of the perpendicular arrangement of the $Pu(PW_{11})_2$ complexes, which minimizes interactions between the complexes and leads to more isolated Pu(IV) ions.

At the junction of the two perpendicularly arranged $Pu(PW_{11})_2$ complexes is one Cs^+ counterion that bridges three $PW_{11}O_{39}{}^{7-}$ POM ligands, two via the equatorial plane of one $Pu(PW_{11})_2$ complex plus one via the tip of the other $Pu(PW_{11})_2$ complex (Figure 6A). This Cs^+ counterion coordinates to 7 oxygens from the POMs (average Cs−O bond of 3.27 Å) plus one water molecule (Cs−O bond of 3.62 Å). This specific Cs^+ counterion is located only 4.45 Å from the nearest Pu^{4+} cation (Pu2 site) ion. The presence of this counterion seems particularly crucial to stabilize the arrangement of the $Pu(PW_{11})_2$ complexes in the lattice as well as efficiently promoting crystallization, crucial for handling and manipulating minute quantities of transuranics.

Each Pu site features three Cs^+ counterions located in the equatorial plane of their respective $Pu(PW_{11})_2$ complex. The Cs−Pu distances are remarkably short (averages of 4.58 and 4.67 Å for the Pu1 and Pu2 sites). Such a close proximity between the two cations is reminiscent of what we recently observed for a smaller Pu polyoxotungstate compound, $Cs_8[Pu(W_5O_{18})_2]{\cdot}CsCl{\cdot}6.5H_2O$, where the central Pu(IV) cation is surrounded by five Cs^+ counterions at 4.35−4.67 Å.

When comparing the three Pu(IV) polyoxotungstate complexes that have been structurally characterized so far (Table 2), we note that $Pu(PW_{11})_2$ exhibits properties that are between that of its analogous complexes with $W_5O_{18}{}^{6-}$ and $P_2W_{17}O_{61}{}^{10-}$. The three polyoxotungstate compounds offer somewhat of a gradient that could be used to manipulate the properties of the Pu(IV) ion in terms of bond distances and distortion from ideal symmetry. When focusing on the first coordination sphere of Pu, we observe a shortening of the Pu−O bonds as the POM gets larger (2.38 Å for $W_5O_{18}{}^{6-}$, 2.35 Å for $PW_{11}O_{39}{}^{7-}$, and 2.32 Å for $P_2W_{17}O_{61}{}^{10-}$). While the $[Pu(P_2W_{17}O_{61})]^{16-}$ complex was isolated as a potassium salt and not a cesium salt like the other two (Table 1),[32] we note that the number of alkali counterions in the vicinity of Pu(IV) decreases when the average Pu−O distance gets shorter, from five Cs^+ with $W_5O_{18}{}^{6-}$, to three Cs^+ with $PW_{11}O_{39}{}^{7-}$, and just one K^+ with $P_2W_{17}O_{61}{}^{10-}$ (Table 2). This trend seems counterintuitive as the overall charge of the Pu−POM complex gets higher, and more electron density is available to counterbalance the Pu cation. The modest vertical squeeze of the complex around Pu (shortening of the Pu−O distance by ~0.06 Å) seems to have a significant impact on the positions of the alkali counterions. This highlights the strong interplay in these compounds among the actinide−POM interactions, alkali−POM interactions, and actinide−alkali interactions.

DISCUSSION

Comparison of the M(IV) Keggin Complexes with Their Oxides MO_2. As we previously synthesized the analogous Keggin compounds with Zr(IV), Hf(IV), Ce(IV), Th(IV), and Np(IV) and determined their structures,[42,47] we can benchmark the newly obtained $Cs_{10}Pu(PW_{11})_2$ compound. Figure 7 compiles the average M^{IV}−O bond lengths in the series of $M^{IV}(PW_{11}O_{39})_2$ compounds. The correlation between the ionic radius of the central cation is strongly linear. Despite the diversity of lattice arrangements that these POM complexes exhibit (e.g., perpendicular versus parallel arrangements), the first coordination sphere of the central cation is highly predictable and seems to be dependent mainly on the ionic radius of the central cation.

This correlation is reminiscent of that observed in the related fluorite-type oxides. In Table S2, we compiled the metal−oxygen bond distances derived from previously published MO_2 oxides for Zr(IV), Hf(IV), Ce(IV), and for all the known tetravalent actinides: Th(IV), Pa(IV), U(IV), Np(IV), Pu(IV), Am(IV), Bk(IV), and Cf(IV). Figure S18 shows that there is a strictly linear correlation between the 8-coordinated ionic radius

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of the tetravalent cations and the metal−oxygen bond distances experimentally determined in MO_2 from powder XRD. The Pu−O distance in PuO_2 is 2.34 Å[55] and $Pu(PW_{11})_2$ offers the closest analog to this among the Pu−POMs (Table 2). The coordination environment of the Pu(IV) ion in the $Pu(PW_{11})_2$ complex resembles its coordination in PuO_2: both are 8-coordinate, the number and nature of neighboring atoms are the same, and the Pu−O bond distances are nearly identical. The main difference is that the POM complex plutonium is antiprismatic, whereas the PuO_2 is prismatic or cubic (Figure 8).

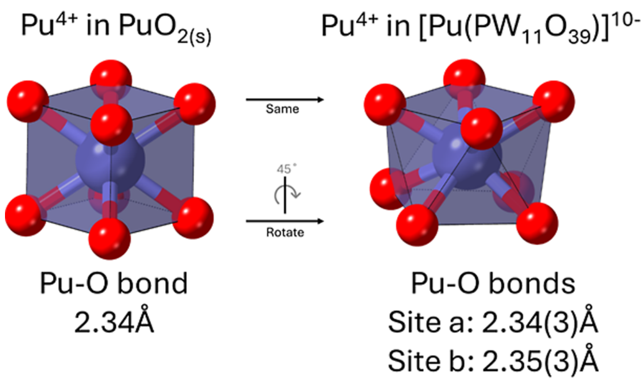

Figure 8. Comparison of the coordination sphere of Pu(IV) in PuO_2 and in $Cs_{20}[Pu(PW_{11}O_{39})_2]_2{\cdot}13H_2O$. The oxide PuO_2 provides a cubic environment.[55] The Pu(IV) complex with the $PW_{11}O_{39}{}^{7-}$ Keggin ion provides a square antiprismatic environment. Two Pu sites are present in the unit cell of the POM complex. See Table S2 for more information. The numbers in parentheses correspond to one standard deviation. Color code: red O, purple Pu.

The similarity between PuO_2 and $Pu(PW_{11})_2$ extends to the series of tetravalent cations, and there is an almost perfectly linear correlation between the M−O bond distances observed in the MO_2 oxides and the $M(PW_{11})_2$ complexes with the tetravalent cations. As shown in Figure 9, there is a clear empirical correlation between the M(IV)−O bond distances in the Keggin POM complexes and those in the corresponding dioxides. Given the low amount needed to crystallize POM compounds, the microscale approach could be used to synthesize Keggin POM complexes with even more exotic tetravalent ions (e.g., Pa(IV), Pr(IV), Tb(IV), Am(IV), Bk(IV)...) and manipulate the properties of these ions in discrete molecules (in solution and/or solid state) rather than in extended solids such as oxides.

■ CONCLUSIONS

This study reports the first Keggin complex containing a plutonium(IV) ion. This new Pu−POM compound was isolated as $Cs_{20}[Pu(PW_{11}O_{39})_2]_2{\cdot}13H_2O$, using our recently developed microscale POM synthesis. Taken together with the only other three Pu(IV)−POMs previously known (with the POMs $W_5O_{18}{}^{6-}$, $P_2W_{17}O_{61}{}^{10-}$, and $Mo_{12}O_{42}{}^{12-}$), this study completes the series of Pu(IV)−POM. The particular chemistry of POMs relative to traditional ligands used in Pu chemistry, with notably their interactions with alkali counterions, allows for fine-tuning the geometry around the Pu cations. When compared to Zr(IV), Hf(IV), Ce(IV), Th(IV), and Np(IV), which also form 1:2 complexes with the $PW_{11}O_{39}{}^{7-}$ POM ligand, Pu(IV) exhibits a similar first coordination sphere. However, the $Pu(PW_{11})_2$ compound departs from the other tetravalent cations by the long-range arrangement of its complexes relative to one another.

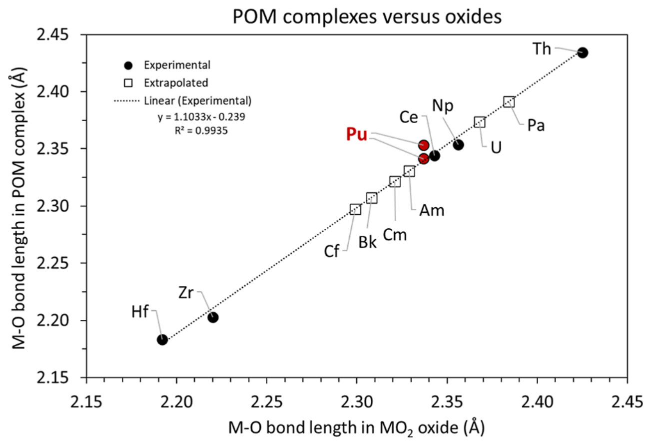

Figure 9. Comparison of the average metal−oxygen bond distances in POM complexes of tetravalent cations ($A_{10}M^{IV}(PW_{11}O_{39})_2{\cdot}xxH_2O$; A = monovalent counterion) versus the corresponding oxides ($M^{IV}O_2$). The circles indicate experimental values for both the POM and oxide bond distances. The squares indicate cases where only the oxide has been experimentally characterized.[55−64] We here extrapolated M−O distances for the $[M^{IV}(PW_{11}O_{39})_2]^{10-}$ complexes with M = Cf^{4+}, Bk^{4+}, Cm^{4+}, Am^{4+}, U^{4+}, and Pa^{4+}. See Table S3 for numerical values. See Figure S18 for the linear correlation between the ionic radii of the cation and the M−O bond distances in MO_2 structures.

The $Pu(PW_{11})_2$ complexes organize perpendicular to each other in the lattice, while the similar complexes with the other tetravalent cations remain parallel. This highlights the uniqueness and unpredictability of Pu chemistry. This study also demonstrates that the $M^{IV}(PW_{11})_2$ complexes display a very similar first-shell coordination chemistry as the corresponding $M^{IV}O_2$ oxides. The results indicate that our microscale POM approach could be a viable pathway to harness the physicochemical properties of rare actinide ions (e.g., Bk^{4+}, Am^{4+}, Pa^{4+}) in discrete molecules, beyond the traditional oxide extended solids. Work is underway to extend the present work to other metal−POM systems.

■ MATERIALS AND METHODS

Caution! *Weapons-grade plutonium (^{wg}Pu) and ^{232}Th, as well as their decay products, constitute serious health hazards because of their radioactive and chemical properties. Radiochemical experiments with Pu were conducted at Lawrence Livermore National Laboratory and Sandia National Laboratories in facilities designed for the safe handling of short- and long-lived radioactive materials and associated waste.*

Materials. A working stock solution of Pu(IV) in HCl was prepared from a primary source of ^{wg}Pu from the LLNL inventory. The isotopic composition was 94% ^{239}Pu, 6% ^{240}Pu, <0.1% ^{241}Pu, <0.1% ^{242}Pu, and <0.05% ^{238}Pu. Nonradioactive chemicals ($Na_2WO_4{\cdot}2H_2O$, $NaCH_3COO{\cdot}3H_2O$, CsCl, $ZrCl_4$, $HfCl_4$, $Ce(NH_4)_2(NO_3)_6$, and $Th(NO_3)_4{\cdot}4H_2O$) were purchased from chemical providers (VWR, Millipore Sigma, and Strem Chemicals) and used as received. All solutions were prepared using deionized water purified by a reverse osmosis cartridge system (≥18.2 MΩ·cm). All experiments were performed in a temperature-controlled room (22 °C) and under an air atmosphere.

Synthesis. POM solutions were made fresh for each experiment. PW_{11} was formed from the in situ conversion of a precursor, $Na_9PW_9O_{34}{\cdot}16H_2O$ (PW_9) in 0.1 M $NaCH_3COO$ at pH 4.5−5.5.[65] PW_9 was synthesized in bulk according to the procedure previously reported by Contant et al.[65] For the title compound ($Cs_{20}[Pu(PW_{11}O_{39})_2]_2{\cdot}13H_2O$), 250 μL of an aqueous solution containing 0.1 mM Pu and 0.2 mM POM was prepared from the Pu(IV) and PW_{11} stock solutions mentioned above. These 250 μL contained 6.0 μg of Pu. To this sample, 20 μL of 6 M CsCl was added. The sample was left at room temperature, and pale red-orange crystals appeared after 2 to 5 days. This microscale synthesis of $Pu(PW_{11})_2$ was repeated at least three times, with similar results, based on the crystal morphology, Raman analysis, and solid-state UV−vis absorbance. Two batches of crystals were also analyzed via single crystal XRD and showed the same unit cell. For the equivalent Zr, Hf, Ce, and Th compounds, the synthesis was done similarly but with larger samples (1 mL instead of 250 μL and 0.5 mM metal). The solutions were also buffered with $NaCH_3COO$ and crystals formed by the addition of CsCl.

Solution-State UV−Visible−NIR Spectrophotometry. Absorbance spectra of the aqueous samples were measured by using a high-performance Cary 6000i UV−vis−NIR spectrophotometer (Agilent Technologies). Samples were contained in sealed quartz cuvettes with a path length of 10 mm. Spectra were corrected blank by measuring the absorbance of the corresponding buffer prior to each titration.

Solid-State UV−Visible Spectrophotometry. Absorbance spectra of the solid samples were measured by using a CRAIC Technology 508 PV microspectrophotometer. Crystals were isolated from their mother liquor and the analysis was performed on at least six crystals. All crystals gave similar spectra. The spectral window was 400−800 nm as accessible by the white light source of the instrument. The reported spectrum is an average of 600 scans with an integration time per scan optimized by the instrument software for each sampling area.

Raman Microscopy. Raman spectra were collected using a Senterra II confocal Raman microscope (Bruker), equipped with high-resolution gratings (1,200 lines/mm), a 532 nm laser source (operated at 15 mW), and a TE-cooled CCD detector. Reported spectra are the average of at least 5−10 different spots per sample, each spot analysis consisting of 64 scans. Multiple crystals of the studied compound were analyzed to confirm the reproducibility of the results. The integration time was set to 100 ms per scan. The sampling area was visually inspected using a microscope camera before and after Raman data collection. No damage to the sample was observed due to laser irradiation.

Small-Angle X-ray Scattering (SAXS). Aqueous samples containing 5 mM cation and 10 mM POM were prepared in 0.1 M $NaCH_3COO$ buffer at pH 5. SAXS data were collected on an Anton Parr SAXSess instrument utilizing Cu Kα radiation and line collimation. Data was recorded on an image plate in the range of 0.08−2.5 Å^{-1}. The sample−plate distance is 26.1 cm. Samples were measured in 1.5 mm glass capillaries (Hampton Research) sealed with wax. Water was used as a background, and scattering was measured for 30 min. SAXSQUANT software was used for data collection and processing (normalization, primary beam removal, and background subtraction). Pair distance distribution function (PDDF) analysis of the scattering data was carried out utilizing Moore's indirect Fourier transform[66] in the IRENA30 macros within IGOR Pro.[67] Simulated scattering curves of $Ce(PW_{11})_2$, $Th(PW_{11})_2$, $Zr(PW_{11})_2$, $Hf(PW_{11})_2$, PW_{12}, and P_2W_{18} were generated using SolX[48] utilizing structural files[39] of $Nd(PW_{11})_2$ (.xyz) (adjusting the central metal site accordingly for M = Zr, Hf, Th, Ce), PW_{12} (.xyz),[68] and P_2W_{18} (.xyz).[35] The solutions were prepared by combining (with stirring) acidic solutions of the metal salt to a solution of PW_{11} (formed in situ from a PW_9 precursor at pH 4.5−5.5)[69] in 0.1 M $NaCH_3COO$ at pH 5. The solutions were filtered using a syringe with a 0.45 μm nylon filter prior to measurement to remove any solid particles.

^{31}P NMR Spectroscopy. An aqueous sample containing 0.3 mM Pu and 0.6 mM PW_{11} was prepared in 2.0 M $NaCH_3COO$ buffer adjusted to pH 4.6. The sample's solvent was 10% D_2O and 90% H_2O. The sample volume was 250 μL, which corresponds to a mass of ^{239}Pu of 18 μg. The $Pu(PW_{11})_2$ sample was loaded into a 4 mm PTFE liner and nested inside an NMR tube for radiological containment. We recently reported[47] similar NMR experiments for the corresponding samples with $Zr(PW_{11})_2$, $Hf(PW_{11})_2$, and $Th(PW_{11})_2$. NMR spectra were collected on a 600 MHz Bruker AVIII spectrometer with a Bruker 5 mm BBFO probe. ^{31}P 1D spectra were collected using a 90° pulse (12.10 μs) with a 4 s recycle delay. 1024 scans were collected at 298 K. All ^{31}P spectra were internally referenced to the ^{2}H lock signal of the 90%/10% solution of H_2O and D_2O.[70]

Crystallography. The structure was collected using a Rigaku Synergy-S single crystal diffractometer, equipped with a kappa goniometer, and using Mo Kα radiation (λ = 0.71073 Å) with a fwhm of ∼200 μm at the sample from a microfocus source. Images were recorded on a Dectris Pilatus 3R (300 K—CdTe) detector and processed using CrysAlis^{Pro}. After integration both analytical absorption and empirical absorption (spherical harmonic, image scaling, detector scaling), corrections were applied.[71] The structure was solved by the Intrinsic Phasing method from the SHELXT program,[72] developed by successive difference Fourier syntheses, and refined by full-matrix least-squares on all F^2 data using SHELX[73] via the OLEX2 interface.[74] Crystallographic information for the reported structure can be obtained free of charge from the Cambridge Crystallographic Data Center (https://www.ccdc.cam.ac.uk/) upon referencing the CCDC number 2476652. Further details of the crystallographic results are given in the Supporting Information.

■ ASSOCIATED CONTENT

Data Availability Statement

The data supporting this article have been included as part of the Supporting Information. The crystal structures of the $Pu(PW_{11})_2$ compound reported here have been deposited in the Cambridge Crystallographic Data Center under the CSD number 2476652. A preliminary version of this study was released as a preprint.[75]

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.inorgchem.5c03627.

Crystallography table, UV−vis analysis, detailed Raman spectra, pair distance distribution function analysis, SAXS analysis, solution-state ^{31}P NMR spectra, additional structure visualizations, bond valence calculations, compilation of bond distances for MO_2 oxides and POM compounds, and references (PDF)

Experimental data for the crystal structure of $Cs_{20}[Pu(PW_{11}O_{39})_2]_2 \cdot 13H_2O$ (CIF)

Accession Codes

Deposition Number 2476652 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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